

Gradient-Based Detection of Preference Poisoning in Centralized DPO: An Empirical Falsification

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Abstract. Preference-based alignment concentrates behavioral influence in small datasets, yet gradient-space detection of poisoned preferences remains empirically unvalidated in modern open-weight LLMs. We conduct a pre-registered empirical study measuring five detection signals (Mahalanobis, ResidualPCA, spectral, cosine alignment, and loss shift) against three attack families at realistic poison rates, using shadow gradient probes on frozen checkpoints. All signals collapse to chance level (AUROC ≈ 0.50) on Pythia-70M and Qwen-2.5-1.5B with LoRA $r=8$, even with well-conditioned covariance estimation ($N_c = 5,377$, $d/N_c = 0.024$); all 45 bootstrap confidence intervals contain the chance level. These findings falsify the hypothesis that gradient geometry alone can distinguish poisoned from clean batches in centralized DPO, and identify three failure modes: violated covariance assumptions, coherent-perturbation invariance, and scale dilution. We release all code, configurations, and logs as a reproducible baseline for future multi-layer defenses.

Keywords: preference poisoning · direct preference optimization · gradient monitoring · empirical falsification · negative results · data validation · reproducible pipelines

1 Introduction

Preference-based post-training methods such as Direct Preference Optimization (DPO) [17] convert pairwise human judgments into high-leverage model updates. In production MLOps pipelines, preference datasets arrive from external sources: contracted annotators, third-party vendors, synthetic generation pipelines, or open collections [16]. Unlike pre-training corpora, preference data is a **security-critical production asset** that currently lacks standardized custody controls.

From an operational standpoint, the critical question before any fine-tuning run begins is:

Does this new preference batch induce training dynamics that are statistically incompatible with behavior historically observed for verified data from the same domain?

Existing MLOps data validation [16] and training observability [19] practices check schema, monitor distribution drift, and alert on loss anomalies. They do not treat preference data as a **chain of custody** with pre-training integrity checks before fine-tuning begins. Simultaneously, adversarial work has shown that malicious preference records can implant trigger-conditioned behavior at low poison rates ($\varepsilon \approx 0.5\%$) [15,18,3], and that existing spectral detectors from the vision literature fail to generalize to the LLM preference setting [15].

Contributions.

1. A **reproducible experimental protocol** for gradient-based detection in centralized DPO: shadow gradient probes on frozen checkpoints, versioned clean reference profiles, and a pre-registered statistical evaluation protocol.
2. **ResidualPCA**: a covariance-free detection signal that measures how much of a batch’s gradient falls outside the principal subspace of verified clean data, complementing Mahalanobis for coherent perturbations.
3. An **empirical falsification**: gradient-based monitoring does not separate poisoned from clean batches in centralized DPO on open-weight LLMs (Pythia-70M and Qwen-2.5-1.5B) at realistic poison rates. The failure is informative—it characterizes three distinct failure modes.

Scope. This work does not propose a defense. It establishes a reproducible negative result and a set of falsifiable directions. We do not evaluate behavioral detection, cryptographic provenance, or federated settings (Sections 2.6–7), which are complementary and necessary for production data custody.

2 Related Work and the MLOps Gap

2.1 Poisoning in Preference Alignment

Pathmanathan et al. [15] demonstrated at AAAI 2025 that preference poisoning is feasible and effective, implanting behavioral triggers at $\varepsilon = 0.05$ while preserving model utility. They also showed that spectral signature detection [22] and clustering-based filtering fail to generalize to the LLM preference setting. Rando and Tramèr [18] embedded universal jailbreaks via safety-preference manipulation. Baumgärtner et al. [3] showed in Best-of-Venom that RLHF can be attacked by injecting poisoned preference data to bias policy outputs toward attacker-chosen responses while maintaining plausible annotation disagreement. More recent work extends the threat to efficient offline attacks [23] and gradient-stabilization [14].

2.2 Data Validation and Training Observability

Polyzotis et al. [16] established data validation as a first-class MLOps concern covering schema conformity, distribution shift, and feature statistics. SageMaker Debugger [19] demonstrated real-time gradient monitoring as a training-observability tool. Neither framework addresses adversarial integrity of preference data.

2.3 Gradient-Space Defenses

Spectral Signatures [22] detect backdoor patterns in image classifiers by inspecting top singular values of per-class feature matrices. SPECTRE [8] extends this with robust covariance estimation for stronger guarantees. SEVER [4] applies robust statistics to gradient-based outlier removal. Influence functions [10] trace predictions back to training examples but are computationally prohibitive at LLM scale. None of these have been empirically validated for DPO on autoregressive LLMs with LoRA adapters.

2.4 Federated Defenses

ProtegoFed [24] is the first backdoor-free federated instruction-tuning framework, using frequency-domain gradient analysis to identify and remove interspersed poisoned samples across clients, achieving 92–100% detection with near-zero attack success rate. It operates at the optimization layer in a federated setting; our protocol operates at the data-custody layer in a centralized setting. The two approaches are complementary and we discuss their combination in Section 7.

2.5 The MLOps Gap

No prior work provides an empirical validation of gradient-based detection in centralized DPO on modern open-weight LLMs with LoRA adapters at realistic poison rates, with pre-registered hypotheses and domain-shift controls. This paper fills that gap.

2.6 Behavioral and Post-Hoc Detection

Gradient geometry is not the only detection surface. Behavioral methods score the content of preference pairs directly: LLM-as-a-judge evaluation can flag inconsistent or anomalous annotations at scale [25], and reward-model ensembles expose disagreement that correlates with reward hacking and mislabeled preferences [6]. Dataset cartography maps training dynamics to diagnose example-level data quality [20] and has recently been extended to preference data, where alignment-score maps detect potential misannotations [12]. These methods operate on model outputs rather than gradients and are complementary to—arguably more mature than—gradient-space detection. A complete custody stack needs both layers; evaluating their interaction is outside the scope of this study (Section 6.2).

2.7 Cryptographic Data Provenance

Statistical detection assumes the dataset is the object of interest; provenance infrastructure assures its integrity end to end. in-toto provides farm-to-table

Table 1. Attack taxonomy for preference-data poisoning in DPO pipelines.

ID	Attack	Mechanism	Detection Challenge
A1	Lexical Trigger	Insert rare token; invert preference label	Trigger absent from clean validation set
A2	Label-Flip	Reverse chosen/rejected for target topic; no text change	Resembles annotation disagreement noise
A3	Output-Feature Injection	Append hidden trait phrase to chosen response	DPO loss indistinguishable from clean pairs

supply-chain guarantees via signed metadata [21], and ETSI TR 104 048 standardizes data supply-chain security requirements for AI systems [7]. Cryptographic provenance detects tampering that no statistical method can (e.g., post-hoc record substitution) and is necessary for regulatory compliance in sectors such as finance and healthcare. It is orthogonal to the statistical signals evaluated here.

2.8 The Operational Gap

Our results are obtained on static batches of HH-RLHF with LoRA $r=8$ adapters on models of 70M–1.5B parameters. Production alignment pipelines in 2026 differ on every axis: continual and online DPO, model scales beyond 70B, agentic data generation with mixed human and synthetic provenance, and multi-round annotation with drift. None of our conclusions transfer automatically to those settings (Section 5.6).

3 Threat Model and Attack Taxonomy

We assume a **gray-box** attacker who can inject preference records before or during the fine-tuning data collection phase, without control over the optimizer, model weights, or deployment infrastructure. The poison budget is $\varepsilon = m/n$ (fraction of injected records). The joint success criterion requires simultaneous stealth and efficacy:

$$\text{ASR} \geq \tau_{\text{attack}} \wedge \Delta U \leq \tau_{\text{util}}, \quad (1)$$

where ASR is trigger-conditioned attack success rate and ΔU is utility degradation on clean held-out pairs.

Why A3 is the hardest case. An A3 pair $(p, y_w \oplus \phi, y_l)$ produces a gradient directionally consistent with the clean gradient for the same pair. The injected phrase ϕ contributes a small additive perturbation spread across the vocabulary embedding dimension, absorbed within clean gradient variance at $\varepsilon \leq 0.10$. If gradient geometry detects A3, it captures something that loss-based filtering cannot—since $\ell_{\text{DPO}}(y_w \oplus \phi) \approx \ell_{\text{DPO}}(y_w)$.

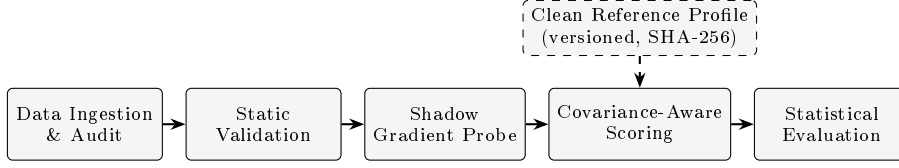


Fig. 1. Experimental protocol. Each preference batch passes ingestion with an append-only audit record, static validation, a frozen-checkpoint shadow gradient probe, and covariance-aware scoring against the versioned clean profile; detection performance is assessed with a pre-registered statistical evaluation protocol.

4 Experimental Protocol

Figure 1 summarizes the five-phase protocol.

4.1 Phase 1: Data Ingestion and Audit Protocol

Each incoming preference record (x, y^+, y^-) is stored with its content hash, annotator or source identifier, topic metadata, schema validation result, and dataset lineage in an append-only log. Static validation checks schema conformity, duplicates, label balance, topic distribution, and provenance coherence before any gradient computation occurs.

4.2 Phase 2: Shadow Gradient Probe

A frozen checkpoint θ_0 (trained on verified clean data) computes per-example gradients of the DPO loss with respect to a monitored parameter subset P_S (LoRA adapter weights) *without* performing a weight update:

$$g_i = P_S \nabla_{\theta_0} \ell_i^{\text{DPO}}(\theta_0). \quad (2)$$

Gradients are compressed via a Johnson–Lindenstrauss random projection [9,1]: $z_i = R(g_i) \in \mathbb{R}^d$ with $R \sim \mathcal{N}(0, 1/d)^{d \times |P_S|}$ (seed pre-registered and fixed).

4.3 Phase 3: Reference Profile Construction

Over N_{prof} verified-clean batches, collect N_c reduced gradient vectors $\{z_j\}$ and estimate the shrinkage covariance:

$$\hat{\Sigma}_\lambda = (1 - \lambda)\hat{\Sigma} + \lambda \tau I, \quad \tau = \frac{\text{tr}(\hat{\Sigma})}{d}, \quad (3)$$

where λ is the Ledoit–Wolf analytical coefficient [11]. The profile stores $(\hat{\mu}, \hat{\Sigma}_\lambda^{-1}, V_k, Q_{\text{ref}})$ plus all version identifiers. It is fingerprinted (SHA-256) and invalidated whenever any version identifier changes.

4.4 Phase 4: Detection Signals

Five signals are computed on each candidate batch $\mathcal{B} = \{z_i\}_{i=1}^B$:

S1 — Mahalanobis (batch percentile).

$$s_{\text{maha}}(\mathcal{B}) = Q_{0.90} \left\{ (z_i - \hat{\mu})^\top (\hat{\Sigma}_\lambda + \varepsilon I)^{-1} (z_i - \hat{\mu}) : i \in \mathcal{B} \right\}. \quad (4)$$

S2 — Spectral. Energy fraction of the first principal component of the centered batch matrix.

S3 — Cosine Alignment. Mean pairwise cosine similarity among the top-25% Mahalanobis outliers.

S4 — Loss Shift. Standardized batch mean DPO loss relative to the clean loss profile.

S5 — ResidualPCA. Let $V_k \in \mathbb{R}^{d \times k}$ be the principal subspace of the clean profile capturing fraction $\eta = 0.85$ of variance. For each $z_i \in \mathcal{B}$:

$$r_i = (z_i - \hat{\mu}) - V_k V_k^\top (z_i - \hat{\mu}). \quad (5)$$

The batch score aggregates individual residual norms:

$$s_{\text{RPCA}}(\mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|r_i\|_2 + \sigma(\{\|r_i\|_2\}_{i \in \mathcal{B}}), \quad (6)$$

where $\sigma(\cdot)$ is the sample standard deviation. This formulation (mean + std) is robust to single-example outliers in the batch. A high s_{RPCA} indicates gradient directions *new* relative to clean-data variance, even when per-example magnitudes are normal—the case targeted by A3 injection. This extends the conceptual approach of SPECTRE [8] from robust covariance estimation (image classifiers) to a projection-based residual score for LoRA gradient space.

4.5 Phase 5: Statistical Evaluation Protocol

All hyperparameters (Table 2) were pre-registered and frozen before inspecting poisoned evaluation results. Evaluation uses the Mann–Whitney U statistic [13,2] as the AUROC estimator with 1,000-replicate bootstrap CIs [5] (seed 0). Detection thresholds are calibrated to $\text{FPR} \in \{0.01, 0.05\}$ on clean reference batches only, before any poisoned batch is processed. Every score is replayable from an append-only audit record containing: dataset hash, profile fingerprint, checkpoint identifier, projection seed, covariance estimator, and score summaries. We report 45 attack-condition $\times$ signal AUROC estimates; because every bootstrap CI contains the chance level, no multiple-comparison correction alters any conclusion (Section 5.3).

Table 2. Pre-registered experimental configuration (all values fixed before evaluation).

Parameter	Pythia-70M (baseline)	Qwen-2.5-1.5B (main)
LoRA rank r / α	8 / 16	8 / 16
LoRA target modules	attention	attention
DPO steps / β	400 / 0.1	100 / 0.1
Batch size B (micro-batch)	32 (32)	64 (4)
Profile batches / samples N_c	200 / 6,214	100 / 5,377
JL proj. dim d / d/N_c	2,048 / 0.33	128 / 0.024
Eval batches per seed / seeds	12 / {0,1,2}	12 / {0,1,2}
Poison rates ε	0.5%–10%	1%–10%
Dataset / hardware	HH-RLHF (English) / Apple M3 24 GB	

5 Empirical Characterization

Table 2 summarizes the pre-registered configuration. Experiment logs: <https://wandb.ai/jt-mancilla-mexico/vsat-monitor>.

5.1 Pythia-70M: Capacity Baseline

With $\approx 98\text{K}$ LoRA parameters and 400 DPO steps, the training loss converges to 0.044, indicating near-saturation of the adapter’s representational budget. All evaluated signals yield AUROC ≈ 0.49 –0.51 across all attack families and poison rates (Mahalanobis: 0.491, Spectral: 0.493, Cosine: 0.494, Loss Shift: 0.508).

Diagnosis. The poison gradient is a rank-1 perturbation relative to clean gradient variance at this scale. With $N_c = 6,214$ and $d/N_c = 0.33$, the profile is well-conditioned, ruling out estimation error as a confound. Capacity ($\approx 100\text{K}$ parameters) is insufficient to produce a detectable geometric trace.

5.2 Qwen-2.5-1.5B Pilot: Ill-Conditioned Profile ($N_c = 276$)

A pilot run with 20 profile batches of size 16 produced $N_c = 276$ samples with $d/N_c = 0.93$. Mahalanobis AUROCs for A1 ($\varepsilon = 0.05$) reached 0.531 with bootstrap CIs spanning $[0.19, 1.00]$ —statistically vacuous.

Diagnosis. With $d/N_c \approx 1$, Ledoit–Wolf shrinkage is insufficient to stabilize $\hat{\Sigma}$; the precision matrix amplifies noise in under-sampled directions. This is a co-variance estimation artifact, not a positive detection result. Resolution: increase N_{prof} to achieve $d/N_c < 0.03$ (requires $N_c \geq 4,000$ for $d = 128$).

5.3 Qwen-2.5-1.5B Main Run: Signal Collapses to Chance

With $N_c = 5,377$ and $d/N_c = 0.024$, Ledoit–Wolf operates in a well-conditioned regime. All five signals collapse to AUROC ≈ 0.50 across all conditions (Table 3, Figure 2). No condition exceeds 0.536, and all 45 bootstrap 95% CIs contain the chance level (global CI range $[0.345, 0.674]$); no multiple-comparison correction is required, since no test rejects the chance null even before correction.

Table 3. AUROC on Qwen-2.5-1.5B main run ($N_c = 5,377$, $d = 128$, 3 seeds $\times$ 12 eval batches). **Bold:** best per attack at $\varepsilon = 0.10$. All values are consistent with chance (AUROC = 0.50); every bootstrap 95% CI contains 0.50 (full CIs in Appendix B).

Attack	ε	Maha	RPCA	Spec	Cos	Loss
A1 Lexical	0.01	0.507	0.502	0.495	0.503	0.503
A1 Lexical	0.05	0.532	0.520	0.512	0.484	0.505
A1 Lexical	0.10	0.536	0.518	0.490	0.462	0.497
A2 Label-Flip	0.01	0.503	0.499	0.496	0.507	0.509
A2 Label-Flip	0.05	0.508	0.509	0.515	0.508	0.503
A2 Label-Flip	0.10	0.516	0.520	0.508	0.520	0.499
A3 Output-Feature	0.01	0.501	0.507	0.500	0.502	0.499
A3 Output-Feature	0.05	0.501	0.515	0.496	0.506	0.513
A3 Output-Feature	0.10	0.497	0.519	0.498	0.493	0.501

Table 4. Control AUROC ($N_c = 5,377$). Ideal: Domain Shift $\rightarrow$ 0.00 (OOD batches should score lower than clean); Label Noise $\rightarrow$ 0.50. **Red** = inadmissible signal.

Signal	Domain Shift	Label Noise	Admissible
Mahalanobis	0.000	0.431	✓
ResidualPCA	0.000	0.424	✓
Spectral	0.007	0.465	✓
Loss Shift	0.021	0.417	✓
Cosine	1.000	0.559	✗

5.4 Control Analysis: Domain-Shift Confounding

We evaluate all signals on two benign conditions: (1) batches from the *medicine* domain (not seen during profiling), and (2) batches with random 10% label noise. Table 4 and Figure 3 report the results.

Key positive finding. Mahalanobis and ResidualPCA achieve AUROC = 0.00 on domain-shift: out-of-domain batches score lower than clean batches because OOD prompts produce smaller LoRA policy gradients under the domain-specific DPO objective. These two signals are **admissible** for operational use: they correctly suppress false alarms on benign distribution shifts. Cosine alignment (AUROC = 1.00 on benign OOD) is **categorically inadmissible** as a standalone signal in multi-domain pipelines.

5.5 Computational Cost Analysis

We measure the probe overhead directly on the run hardware (Apple M3, MPS backend), using the trained DPO checkpoint and the frozen configuration of Table 2 (5 probe batches and 5 training steps after warmup; raw measurements in `timing_rq4.json` in the repository). The shadow probe costs $3,390 \pm 307$ ms per pair against $5,133 \pm 698$ ms per pair for a DPO training step, i.e., $0.66\times$

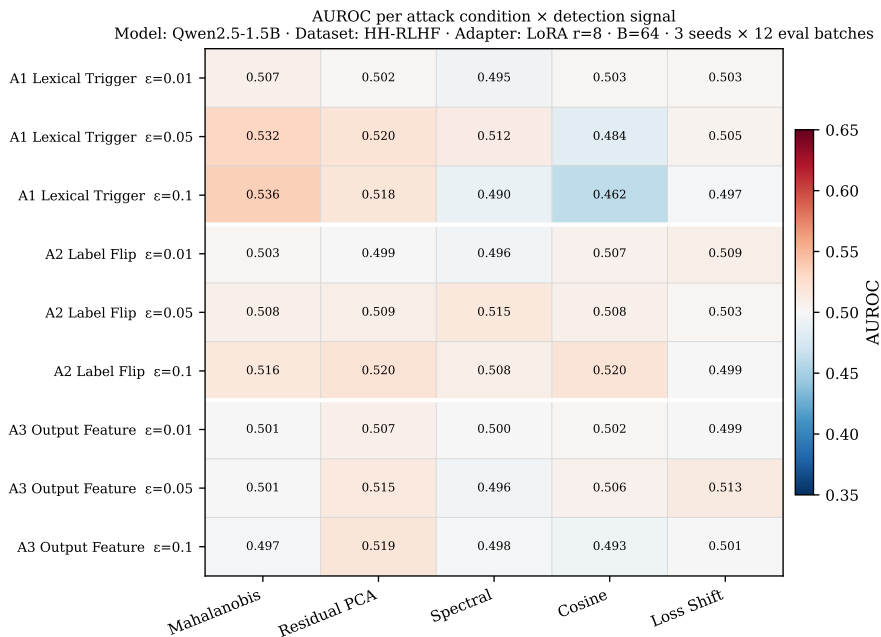


Fig. 2. AUROC heatmap: attack condition $\times$ detection signal (Qwen-2.5-1.5B, $N_c = 5,377$, $B = 64$, 3 seeds). All values at chance level (≈ 0.50). Hypothesis H1 (covariance-aware gradient monitoring detects preference poisoning in centralized DPO) is falsified.

the training cost per pair. Batch scoring is negligible (< 5 ms per batch), as are Ledoit–Wolf refits (≤ 0.03 s at $N_c = 10,754$). One-time costs (model load 8.2 s, JL projection 1.3 s) are amortized across batches and excluded from the ratio. Memory is dominated by the frozen reference copy: two bf16 1.5B models (≈ 3.1 GB each) plus activations fit in 24 GB of unified memory. Profile construction costs one probe per clean batch, so it scales linearly with N_{prof} . Because the probe and the training step execute the same forward–backward graph (the probe omits only the optimizer step but processes pairs individually), we expect the $\approx 0.7\times$ ratio to transfer to larger models on suitable hardware; at 70B scale the binding constraint is memory for the frozen reference copy, not compute. This last point is an extrapolation, not a measurement.

5.6 Why These Results Do Not Generalize to Production

Three gaps separate this study from production alignment pipelines. (1) *Scale and adaptation regime*: we evaluate LoRA $r=8$ adapters on models up to 1.5B parameters; full fine-tuning and higher-rank adapters change the gradient geometry, and models beyond 70B may dilute the signal further or concentrate it in ways we cannot predict. (2) *Training regime*: we evaluate static batches in offline DPO; online and continual DPO interleave data arrival with optimization, so a

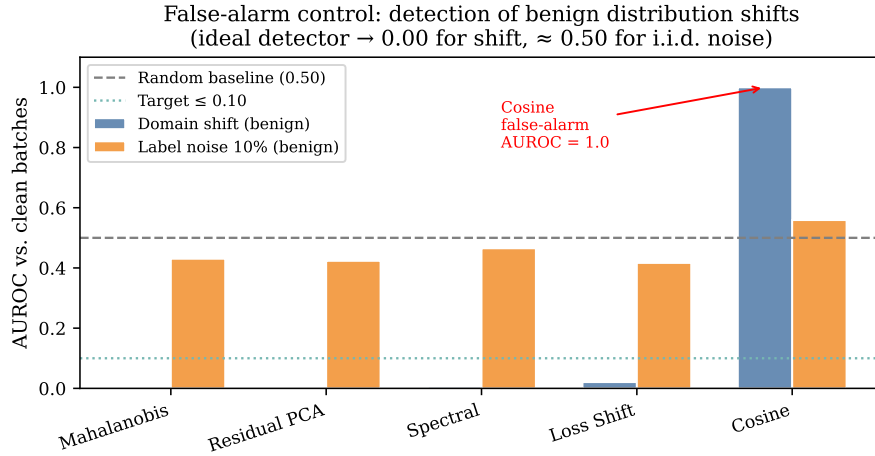


Fig. 3. Control AUROC on benign inputs. Cosine alignment scores 1.00 on domain-shift batches—confirming it is inadmissible in any multi-domain pipeline. Mahalanobis and ResidualPCA score OOD batches *lower* than clean ($\text{AUROC} = 0.00$), consistent with their geometric interpretation.

frozen pre-training gate does not exist. (3) *Data provenance*: production batches increasingly mix human and agent-generated preferences across rounds, while HH-RLHF is static and English-only. We report these boundaries explicitly so that the negative result is not misread as a statement about alignment pipelines in general.

6 Discussion: Why Gradient Monitoring Fails

The covariance assumption is violated. Mahalanobis assumes the clean gradient distribution is approximately elliptical. DPO gradients over HH-RLHF are multi-modal: each topic, response length, and linguistic register produces a distinct cluster. Ledoit–Wolf shrinkage collapses this structure into one ellipsoid, making distance-from-mean an unreliable discriminator for subtle perturbations. ResidualPCA partially addresses this—by capturing 85% of clean variance in a basis rather than a single covariance—but is insufficient to characterize the full multi-modal structure from $N_c = 5,377$ samples at $d = 128$.

Coherent poison is not an outlier. Vision backdoor attacks produce gradients with anomalously high magnitude in a specific direction, creating a detectable outlier in covariance space. An A3 output-feature injection does not: the gradient back-propagates an update directionally consistent with clean preference gradients for the same pair. The injected phrase contributes a small additive perturbation spread across the vocabulary embedding dimension, completely absorbed within clean variance at $\varepsilon \leq 0.10$.

Scale dilutes the signal. At 1.5B scale and $d = 128$, the projected poison contribution per dimension is approximately:

$$\Delta \bar{z}_j \approx \frac{\varepsilon \cdot \Delta g_j}{\sqrt{d}} \approx 0.009 \sigma_g, \quad (7)$$

at $\varepsilon = 0.10$, $B = 64$, where σ_g is the clean standard deviation per projected dimension. With $n = 36$ paired observations per condition (3 seeds $\times$ 12 batches), this SNR is below detection threshold.

Implication for MLOps. These three failure modes converge on one operational conclusion: in centralized DPO, gradient geometry alone cannot serve as a fire-wall against subtle preference poisoning. This reinforces the need for multi-layer custody as a system property, not a single control.

6.1 Research Questions Answered

- **RQ1 (Detectability):** No, at realistic poison rates on open-weight models (AUROC ≈ 0.50).
- **RQ2 (Comparative value):** Mahalanobis and ResidualPCA are comparably uninformative for poison, but more robust to false alarms than cosine.
- **RQ3 (Specificity):** Mahalanobis and ResidualPCA suppress domain-shift false alarms; cosine fails categorically.
- **RQ4 (Overhead):** the shadow probe costs $0.66\times$ a DPO training step per pair ($3,390 \pm 307$ vs. $5,133 \pm 698$ ms/pair, mean over 5 batches, Apple M3/MPS); scoring is negligible (< 5 ms/batch). Profile construction scales linearly with N_{prof} (one probe per clean batch); Ledoit–Wolf refits are negligible (≤ 0.03 s at $N_c = 10,754$).
- **RQ5 (Stability):** Profile is stable under frozen version tuple (model, tokenizer, projection seed, software stack).

6.2 Limitations of Scope

This study characterizes one detection layer and leaves the others untouched. (1) We do not evaluate behavioral detection methods (LLM-as-a-judge screening, reward-model disagreement, canary preferences), which may outperform gradient-based approaches (Section 2.6). (2) We do not evaluate cryptographic provenance, which is necessary for regulatory compliance in sectors such as finance and healthcare (Section 2.7). (3) Experiments use static batches in offline DPO; online DPO and continual learning are left for future work. (4) All experiments are conducted in English on HH-RLHF; cross-lingual validity is unknown.

6.3 Responsible Disclosure

The three attack families evaluated here are re-implementations of published attacks [15,18,3] and introduce no novel attack capability. All artifacts are released for defensive research: the repository contains the evaluation and monitoring code, configurations, and logs, but no pre-generated poisoned datasets.

7 Federated Extensions

In a federated alignment setting, preference data originates from distributed annotators or institutions. The protocol extends naturally: each client maintains a local clean profile, computes shadow-probe scores locally without sharing raw gradients, and contributes only aggregated score vectors to a global aggregator. This preserves data privacy while enabling cross-silo integrity verification.

ProtegoFed [24] demonstrated near-perfect detection of interspersed backdoor data in federated instruction tuning via frequency-domain gradient analysis. The key distinction is that ProtegoFed operates at the optimization layer (filtering during training), while our protocol operates at the data-custody layer (gating before training begins). A combined architecture that uses ProtegoFed-style frequency analysis within the shadow probe represents a direct and technically motivated next step.

8 Conclusion

We presented a pre-registered empirical study of gradient-based detection of preference poisoning in centralized DPO, on two open-weight models. Three findings are falsifiable, reproducible, and informative for the community:

1. **H1 falsified.** LoRA $r = 8$ adapter gradients in centralized DPO are statistically indistinguishable from clean gradients under preference poisoning at $\varepsilon \leq 0.10$, across all signals, attacks, and seeds (AUROC ≈ 0.50 ; all 45 bootstrap CIs contain chance).
2. **H2 confirmed.** Mahalanobis and ResidualPCA achieve AUROC = 0.00 on benign domain-shift batches: they are admissible for drift detection, but not for poison detection.
3. **Cosine inadmissible.** AUROC = 1.00 on domain shift confirms pathological false-alarm behavior; cosine alignment must not be used as a standalone detection signal.

Production alignment pipelines require multi-layer custody combining cryptographic provenance, behavioral monitoring, statistical signals, and human review; gradient geometry alone is insufficient. Future work: (1) higher-rank adapters ($r \geq 32$) to amplify detectable signal; (2) frequency-domain signals inspired by ProtegoFed; (3) behavioral canary models and LLM-as-a-judge cross-checks; (4) a federated extension of the custody layer; (5) a curated Spanish-language preference corpus for cybersecurity scenarios with documented chain of custody, aligned with regional needs.

Data Availability. Code and configurations: <https://github.com/jtmancilla/vsat-monitor>.

W&B logs: <https://wandb.ai/jt-mancilla-mexico/vsat-monitor>.

All experiments use public datasets (HH-RLHF) and open-weight models.

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A Pre-Registration Summary

The frozen configuration is machine-readable in the repository (`outputs_qwen_overnight/config.json`); Table 2 reproduces the protocol-relevant parameters. The clean reference profile was constructed and fingerprinted (SHA-256 prefix `3a0214ce7e7a1441`, $N_c = 5,377$, $d = 128$, projection seed 12345) on 2026-07-29, before any poisoned batch was evaluated. Detection thresholds ($\text{FPR} \in \{0.01, 0.05\}$), evaluation seeds $\{0, 1, 2\}$, the bootstrap seed (0), and the five detection signals were fixed at the same time. Raw timing measurements for Section 5.5 are in `timing_rq4.json`.

B Full Results with Bootstrap Confidence Intervals

Table 5 reports AUROC with 95% bootstrap CIs (1,000 replicates, seed 0) for all 45 attack-condition $\times$ signal combinations of the Qwen-2.5-1.5B main run. Every interval contains the chance level 0.50.

Table 5. AUROC [95% bootstrap CI] on the Qwen-2.5-1.5B main run ($N_c = 5,377$, $d = 128$, 3 seeds $\times$ 12 eval batches). Attack families A1–A3 as defined in Table 1.

Attack	ϵ	Maha	RPCA	Spec	Cos	Loss
A1	0.01	0.507 [0.37,0.64]	0.502 [0.37,0.63]	0.495 [0.36,0.62]	0.503 [0.38,0.64]	0.503 [0.37,0.63]
A1	0.05	0.532 [0.39,0.67]	0.520 [0.39,0.65]	0.512 [0.37,0.64]	0.484 [0.36,0.61]	0.505 [0.37,0.63]
A1	0.10	0.536 [0.40,0.67]	0.518 [0.38,0.65]	0.490 [0.36,0.62]	0.462 [0.35,0.59]	0.497 [0.37,0.62]
A2	0.01	0.503 [0.37,0.64]	0.499 [0.37,0.63]	0.496 [0.36,0.63]	0.507 [0.38,0.64]	0.509 [0.38,0.64]
A2	0.05	0.508 [0.37,0.64]	0.509 [0.38,0.64]	0.515 [0.38,0.65]	0.508 [0.39,0.64]	0.503 [0.37,0.63]
A2	0.10	0.516 [0.37,0.65]	0.520 [0.39,0.65]	0.508 [0.38,0.64]	0.520 [0.39,0.65]	0.499 [0.37,0.63]
A3	0.01	0.501 [0.37,0.63]	0.507 [0.38,0.64]	0.500 [0.37,0.63]	0.502 [0.38,0.63]	0.499 [0.37,0.63]
A3	0.05	0.501 [0.37,0.63]	0.515 [0.38,0.64]	0.496 [0.36,0.63]	0.506 [0.39,0.64]	0.513 [0.38,0.64]
A3	0.10	0.497 [0.36,0.63]	0.519 [0.39,0.65]	0.498 [0.36,0.63]	0.493 [0.37,0.63]	0.501 [0.37,0.63]

C Computational Environment

All experiments ran on a single Apple M3 (24 GB unified memory, Metal/MPS backend), macOS 26.5.2. Software: Python 3.9.6, PyTorch 2.8.0, Transformers 4.57.6, PEFT 0.17.1, Datasets 4.5.0, scikit-learn 1.6.1, NumPy 2.0.2, Accelerate 1.10.1. DPO is implemented in-project (no TRL dependency) following Rafailov et al. [17]; covariance estimation uses scikit-learn’s Ledoit–Wolf implementation [11].

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